POLI 784 - Problem Set #5

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```r
# load packages into session
library("styler")
library("tidyverse")
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.5     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library("glue")
library("rdrobust")
library("rddensity")
library("mvtnorm")
library("ivDiag")
```

```
## Tutorial: https://yiqingxu.org/packages/ivDiag/
```

Question 1

Part A

The estimand in a sharp RDD is the average treatment effect at the cutoff. $\tau_{SRD} = E[Y_i(1) - Y_i(0)|Z_i = c]$. This is a local causal effect, as it captures the effect of the treatment on the outcome for units that are at the cutoff ($Z_i = c$).

The estimand in a Fuzzy RDD is the Local Average Treatment Effect (LATE) at the cutoff, given by: $\tau_{FRD} = \frac{\mu_{Y+} - \mu_{Y-}}{\mu_{D+} - \mu_{D-}}$. This captures the causal effect for compliers at the cutoff, – those whose treatment status is changed by crossing the threshold.

Part B

In a sharp RDD, we account for the influence of confounders through the design rather than by controlling for them directly. The key identification assumptions are no self-selection around the cutoff and continuity at the cutpoint. Put differently, units just above and just below the threshold must be comparable in all relevant respects except for treatment assignment, and nothing else that affects the outcome can jump discontinuously at the cutoff. Under these assumptions, the difference in outcomes at the cutoff can be interpreted as the causal effect of treatment. Covariates are therefore not required for identification, though they can sometimes improve efficiency.

Part C

What consequence does fitting global polynomials to estimate the causal effect in an RDD have and what alternative estimation strategy is available to us under continuity?

Fitting global polynomials in an RDD is problematic because estimates can be driven by observations far from the cutoff, leading to misleading estimates of the treatment effect. Under the continuity framework, the goal is to approximate the conditional expectation function (CEF) at the cutpoint, but global polynomials impose a single functional form over the entire support of the running variable, which can distort this approximation. Instead, we use local linear regression within a bandwidth around the cutoff to approximate the CEF on each side of the threshold. By focusing on observations close to the cutpoint, this approach provides a more reliable estimate of the causal effect.

Part D

Relying on cross-validation to select the bandwidth h in an RDD is problematic because it chooses h to minimize prediction error, not to minimize the mean squared error of the treatment effect estimator at the cutoff. As a result, the selected bandwidth can be too large, so bias remains non-negligible relative to the standard error. To address this, we can use MSE-optimal bandwidth selectors (Imbens and Kalyanaraman 2012) and apply bias correction. Calonico, Cattaneo, and Titiunik (2014) propose a bias-corrected estimator that uses an auxiliary bandwidth b to estimate and subtract the leading bias term, while adjusting inference accordingly. In practice, we can use rdrobust, which selects both bandwidths automatically.

Question 2

Here, we replicate the main findings of a study by Goyal and Sells (2023) examining how the election of female mayors in Brazil affects party building during their term. The authors use a regression discontinuity design (RDD) to estimate the causal effect of a female candidate winning an election on the gender distribution of recruited party members. Their forcing variable `Margin` is the margin of victory of female mayoral candidates, while their outcome variable `Gender.Gap` is the difference in the recruitment rate of men and women into the female candidate's party per 1000 eligible voters, defined as follows:

$$GenderGap = 1000 \times \left(\frac{MaleRecruits}{MaleVoters} - \frac{FemaleRecruits}{FemaleVoters}\right)$$

Part A

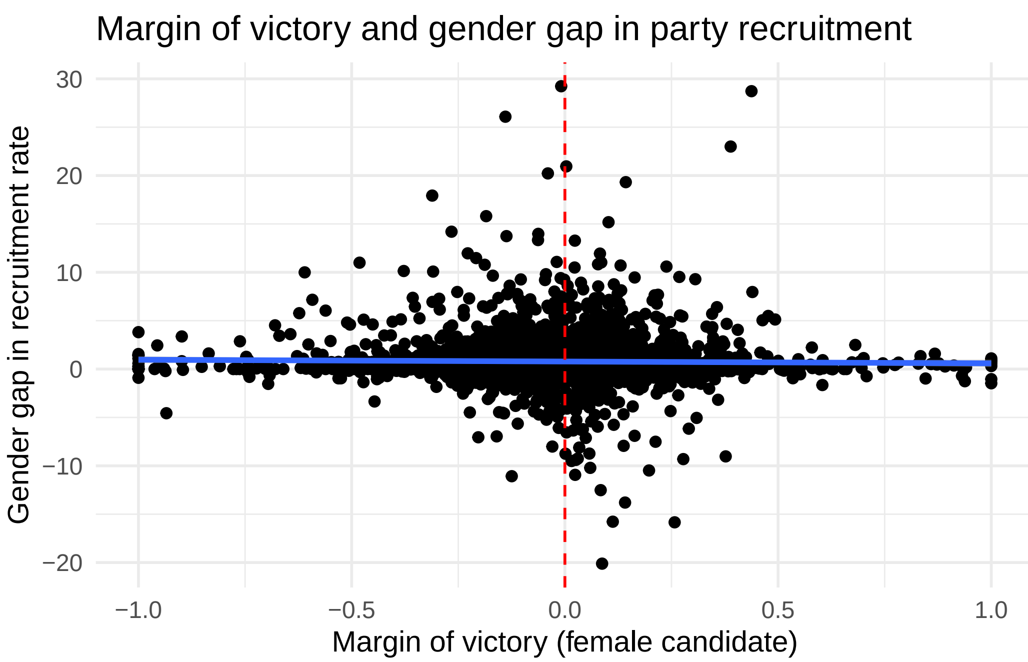
Here, we plot how the outcome variable `Gender.Gap` varies across the forcing variable `Margin`. In this example, "treated" units are municipalities in which the female candidate's party wins the election and "control" units are those in which the male candidate's party wins the election.

```
# Load data
df <- read.csv('data1-1.csv')

# Create scatterplot of outcome vs. forcing variable
ggplot(df, aes(x = Margin, y = Gender.Gap)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  geom_vline(xintercept = 0, color="red",linetype="dashed") +
  labs(
    x = "Margin of victory (female candidate)",
    y = "Gender gap in recruitment rate",
    title = "Margin of victory and gender gap in party recruitment"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 239 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

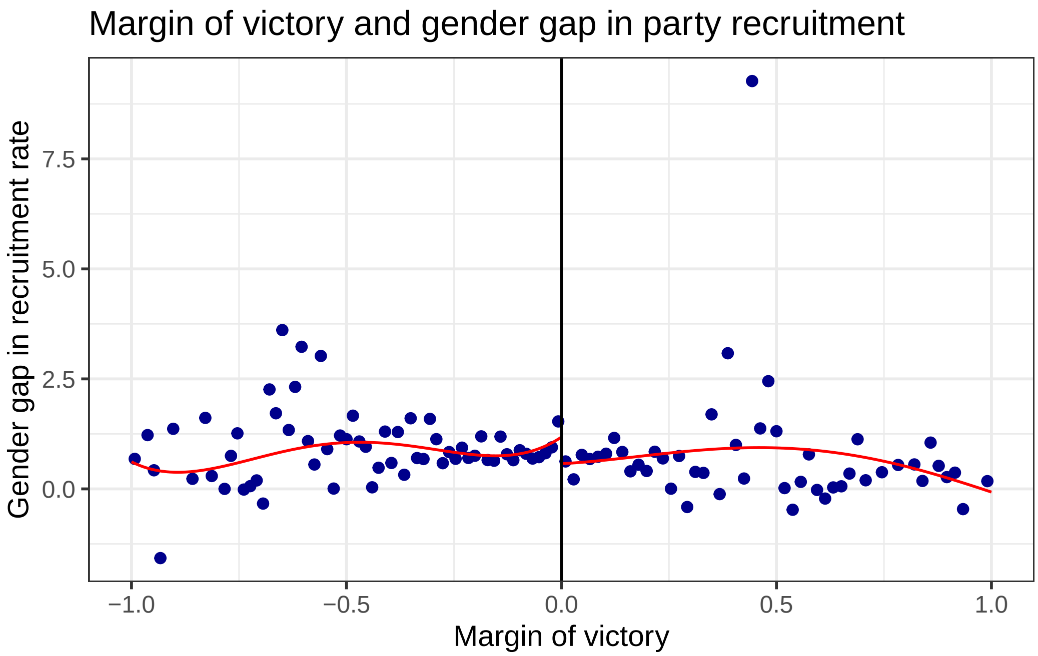
```
Warning: Removed 239 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Part B

Here, we plot the outcome against the forcing variable with a fourth-order polynomial on either side of the RDD cutoff of zero.

```r
# plot cutoff
rdplot(
  y = df$Gender.Gap,
  x = df$Margin,
  c = 0,
  p = 4,
  x.label = "Margin of victory",
  y.label = "Gender gap in recruitment rate",
  title = "Margin of victory and gender gap in party recruitment")
```



In this figure, the red line represents the conditional expectation function (CEF) estimated using a fourth-order polynomial fit, while the blue points represent the local mean outcome within a set of bins that were selected by `rdplot` using a data-driven approach that attempts to mimic the variance of the underlying data. Because each point represents the local mean within each bin, there are far fewer points in this plot than in the one created in Part A.

Part C

Here, we use `rdrobust` to estimate the local average treatment effect (LATE) of a female candidate winning a close mayoral race on the gender gap in the female candidate's party recruitment rate during the subsequent four years.

```
# fit model
fit_rd <- rdrobust(
  y = df$Gender.Gap,
  x = df$Margin,
  c = 0,
  all = TRUE
)

# print results to console
summary(fit_rd)
```

```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 3445
BW type                       mserd
Kernel                   Triangular
VCE method                       NN

Number of Obs.                 1991         1454
Eff. Number of Obs.            1298         1063
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   0.178        0.178
BW bias (b)                   0.393        0.393
rho (h/b)                     0.453        0.453
Unique Obs.                    1973         1445

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional    -0.699     0.238    -2.937     0.003    [-1.165 , -0.232]
Bias-Corrected    -0.790     0.238    -3.321     0.001    [-1.256 , -0.324]
        Robust    -0.790     0.267    -2.960     0.003    [-1.313 , -0.267]
=============================================================================
```

1. The above results imply that in close elections, victory by female candidates reduces the gender gap in their party's recruitment rate by 0.79 recruits per 1000 eligible voters. This reduction is roughly equal to the average gender gap in our sample, which suggests that female mayors can help to overcome existing disparities in party recruitment.
2. In the above model, $h = 0.178$ is the bandwidth used to estimate the LATE, while $b = 0.393$ is the bandwidth used to estimate the bias of the RDD estimator. The `rdrobust` function uses a linear model with bandwidth h to generate a conventional estimate of the LATE and a quadratic model with bandwidth b to estimate how much bias exists in the conventional estimate. This bias estimate is subsequently used to generate the bias-corrected and robust estimates shown above. By default, both steps utilize a triangular kernel.
3. By default, `rdrobust` selects b and h using an approach that seeks to minimize the mean squared error (MSE) of the final estimate. When performing RDD, there is a bias-variance trade off that is influenced by the size of the selected bandwidth: smaller bandwidths have lower bias near the cutoff but higher variance compared to larger bandwidths that include more data from regions further away from the cutoff. Because $MSE \approx Bias^2 + Variance$, the MSE-optimal bandwidth selected by `rdrobust` attempts to strike a balance between the bias and variance of the estimator.

Part D

Here, we use the MSE-optimal bandwidth estimated by `rdrobust` to fit a weighted linear regression model of the outcome on either side of the cutoff.

```r
# MSE-optimal bandwidth
# (by default, rdrobust uses same bw on either side of cutoff)
h <- fit_rd$bws[["h","left"]]

# Cutoff
c <- 0

# Triangular kernel function
triangular_kernel <- function(x, h, c = 0.0) {

  # compute normalized distance from cutoff
  u <- (x - c) / h

  # apply triangular kernel: K(u) = (1 - |u|) if |u| <= 1, else 0
  weights <- ifelse(abs(u) <= 1, 1 - abs(u), 0)

  return(weights)
}

# Apply kernel to forcing variable
df$w <- triangular_kernel(df$Margin,h)


# Split data on either side of cutoff
left_df <- df[df$Margin < c,]
right_df <- df[df$Margin >= c,]

# Fit weighted linear model on either side of cutoff
left_fit <- lm(Gender.Gap ~ Margin, data = left_df, weights = w)
right_fit <- lm(Gender.Gap ~ Margin, data = right_df, weights = w)

# Get intercepts of fitted models
left_intercept <- left_fit$coefficients["(Intercept)"]
right_intercept <- right_fit$coefficients["(Intercept)"]

# Compute LATE as difference in intercepts
LATE <- right_intercept - left_intercept
```

```
# Display to console
glue("LATE(h={round(h,3)}): {round(LATE,3)}")
```

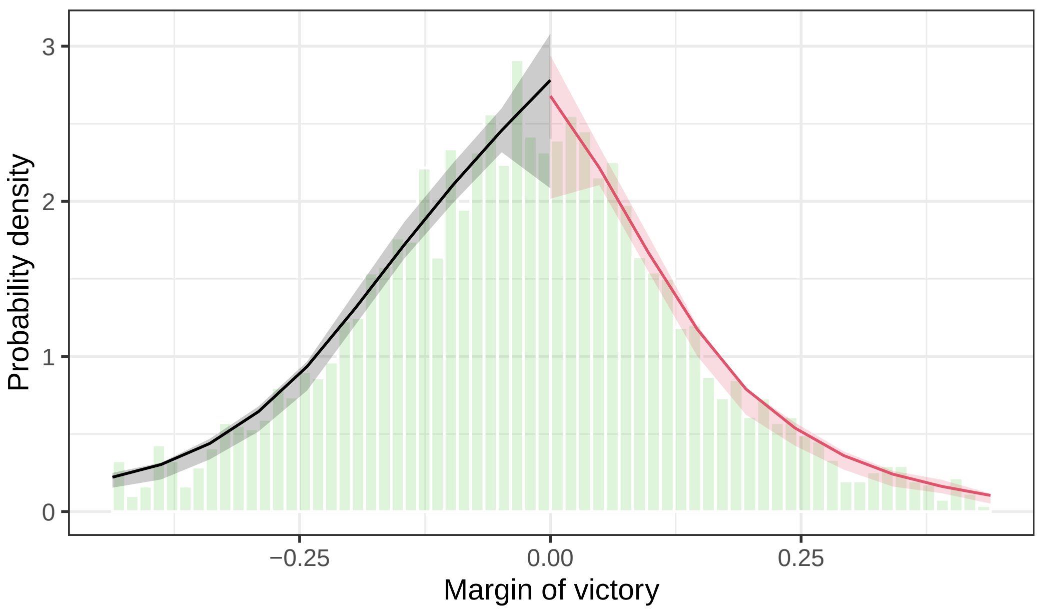
```
LATE(h=0.178): -0.699
```

This result is consistent with the “conventional” (non-bias-corrected) estimate from Part C.

Part E

Here, we use the `rddensity` and `rdplotdensity` functions to test the continuity of the forcing variable across the cutoff.

```
forcing_var_density <- rddensity(X = df$Margin, vce = "jackknife")
p <- rdplotdensity(forcing_var_density,
                   X = df$Margin,
                   xlabel="Margin of victory",
                   ylabel="Probability density")
```



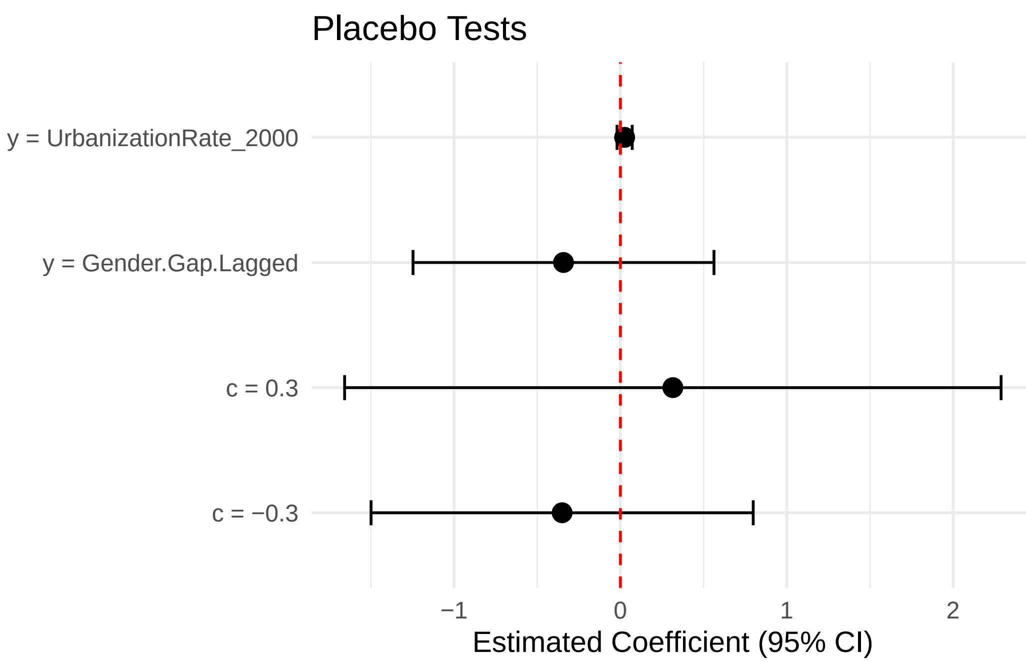
Based on the above plot, there is no evidence of sorting or self-selection around the cutoff.

Part F

Here, we implement several placebo tests to evaluate the robustness of our findings.

```
# Initialize vectors
coef <- rep(NA,4)
LB <- rep(NA,4)
UB <- rep(NA,4)
label <- rep(NA,4)


# Placebo cutoff: c = 0.3
mod <- rdrobust(
  y = df$Gender.Gap,
  x = df$Margin,
  c = 0.3,
  all = TRUE
)

coef[1] <- mod$coef[["Robust","Coeff"]]
LB[1] <- mod$ci[["Robust","CI Lower"]]
UB[1] <- mod$ci[["Robust","CI Upper"]]
label[1] <- "c = 0.3"

# Placebo cutoff: c = -0.3
mod <- rdrobust(
  y = df$Gender.Gap,
  x = df$Margin,
  c = -0.3,
  all = TRUE
)

coef[2] <- mod$coef[["Robust","Coeff"]]
LB[2] <- mod$ci[["Robust","CI Lower"]]
UB[2] <- mod$ci[["Robust","CI Upper"]]
label[2] <- "c = -0.3"

# Placebo outcome: y = Gender.Gap.Lagged
mod <- rdrobust(
  y = df$Gender.Gap.Lagged,
  x = df$Margin,
  c = 0.0,
  all = TRUE
```

```r
)

coef[3] <- mod$coef[["Robust","Coeff"]]
LB[3] <- mod$ci[["Robust","CI Lower"]]
UB[3] <- mod$ci[["Robust","CI Upper"]]
label[3] <- "y = Gender.Gap.Lagged"

# Placebo outcome: y = UrbanizationRate_2000
mod <- rdrobust(
  y = df$UrbanizationRate_2000,
  x = df$Margin,
  c = 0.0,
  all = TRUE
)

coef[4] <- mod$coef[["Robust","Coeff"]]
LB[4] <- mod$ci[["Robust","CI Lower"]]
UB[4] <- mod$ci[["Robust","CI Upper"]]
label[4] <- "y = UrbanizationRate_2000"

# combine vectors into a dataframe
plot_df <- data.frame(
  label = label,
  coef  = coef,
  LB    = LB,
  UB    = UB
)

# create coefficient plot
ggplot(plot_df, aes(x = coef, y = label)) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = LB, xmax = UB), height = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    x     = "Estimated Coefficient (95% CI)",
    y     = NULL,
    title = "Placebo Tests"
  ) +
  theme_minimal()
```



The 95% confidence interval of the LATE estimate includes the null under all four placebo tests. This is reassuring since our study design rests on the assumption that the discontinuity in our outcome is specific to the true cutoff of zero. In addition, we would not expect this cutoff to affect pre-treatment covariates such as the urbanization rate as well as lagged outcomes.

Part G

Here, we evaluate the impact of adding covariates to our model.

```
covariates <- c("University","Married","Incumbent","Age")

mod <- rdrobust(y = df$Gender.Gap,
                x = df$Margin,
                c = 0,
                covs = df[,covariates],
                all = TRUE)

summary(mod)
```

```
Covariate-adjusted Sharp RD estimates using local polynomial regression.

Number of Obs.                 3445
```

```
BW type                          mserd
Kernel                      Triangular
VCE method                          NN

Number of Obs.                    1991         1454
Eff. Number of Obs.               1181         1003
Order est. (p)                       1            1
Order bias  (q)                      2            2
BW est. (h)                      0.157        0.157
BW bias (b)                      0.343        0.343
rho (h/b)                        0.457        0.457
Unique Obs.                       1973         1445

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional    -0.760     0.254    -2.991     0.003    [-1.258 , -0.262]
Bias-Corrected    -0.861     0.254    -3.388     0.001    [-1.359 , -0.363]
        Robust    -0.861     0.285    -3.025     0.002    [-1.419 , -0.303]
=============================================================================
```

In theory, including covariates should improve the efficiency of our estimator. In the above example, the standard errors on our LATE estimates are actually slightly larger when covariates are included; however, the bandwidth h is smaller in the model that includes covariates, which suggests that the estimate obtained using this approach should be less biased.

Part H

Here, we evaluate the impact of making the following modifications to our base model:

1. The selected bandwidth size is allowed to vary on either side of the cutoff.
2. Standard errors are estimated using the HC2 variance estimator.
3. Instead of local linear regression, local quadratic regression is used to construct the point estimator. By the same token, a third-order polynomial is used to construct the bias correction instead of a second-order polynomial.

```
mod <- rdrobust(y = df$Gender.Gap,
                x = df$Margin,
                c = 0,
                bwselect="msetwo",
                vce="hc2",
```

```
                 all = TRUE)

summary(mod)
```

```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 3445
BW type                      msetwo
Kernel                   Triangular
VCE method                      HC2

Number of Obs.                 1991         1454
Eff. Number of Obs.             962          903
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   0.124        0.129
BW bias (b)                   0.233        0.216
rho (h/b)                     0.532        0.598
Unique Obs.                    1973         1445

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional    -0.847     0.287    -2.950     0.003    [-1.410 , -0.284]
Bias-Corrected    -0.973     0.287    -3.388     0.001    [-1.536 , -0.410]
        Robust    -0.973     0.336    -2.899     0.004    [-1.631 , -0.315]
=============================================================================
```

Compared to our original model, the bandwidth h used to construct the point estimate is larger on both sides of the cutoff while the bandwidth b used to construct the bias correction is smaller. This results in a h/b ratio that is much larger than before. This implies that the point estimation and bias correction procedures are using essentially the same subset of the data, which is often undesirable. In addition, the confidence intervals on our estimate of the LATE are much larger than before. This suggests that there may not be a sufficient quantity of data to justify adding complexity to our model by incorporating higher-order polynomials and allowing the bandwidth to vary on either side of the cutoff.

Part I

Here, we test the impact of fitting a fuzzy RD model with a binary treatment indicator that takes on a value of one when the female candidate's margin of victory is greater than or equal to zero.

```r
# Create binary assignment indicator
df$tr <- as.numeric(df$Margin >= 0.0)

# fit fuzzy RD model
mod <- rdrobust(y = df$Gender.Gap,
                x = df$Margin,
                c = 0,
                fuzzy = df$tr,
                all = TRUE)

summary(mod)
```

```
Fuzzy RD estimates using local polynomial regression.

Number of Obs.                 3445
BW type                       mserd
Kernel                   Triangular
VCE method                       NN

Number of Obs.                 1991         1454
Eff. Number of Obs.            1298         1063
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                   0.178        0.178
BW bias (b)                   0.393        0.393
rho (h/b)                     0.453        0.453
Unique Obs.                    1973         1445

First-stage estimates.

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional     1.000     0.000       Inf     0.000     [1.000 , 1.000]
Bias-Corrected     1.000     0.000       Inf     0.000     [1.000 , 1.000]
        Robust     1.000     0.000       Inf     0.000     [1.000 , 1.000]
=============================================================================

Treatment effect estimates.
```

```
========================================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
========================================================================================
  Conventional    -0.699     0.238    -2.937     0.003    [-1.165 , -0.232]
Bias-Corrected    -0.790     0.238    -3.321     0.001    [-1.256 , -0.324]
        Robust    -0.790     0.267    -2.960     0.003    [-1.313 , -0.267]
========================================================================================
```

We see that in this scenario, the fuzzy RD returns the same estimate of the LATE as our original model. This is reassuring since in this example, there are no opportunities for non-compliance (i.e., the candidate with the greater share of the vote is guaranteed to win). This suggests that under full compliance, the fuzzy RD estimator reduces to the sharp RD estimator.

Question 3

Part A

```r
#load data into session
data <- read_csv("data2-1.csv")
```

```
Rows: 1604 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (4): Y, D, Z, cluster

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#i.
#Compute the proportion of always-takers, never-takers, and compliers in the sample. #
# Make sure that these proportions sum to 1 and print the results to console.

#compute the proportion of always-takers
always_takers <- mean(data$D[data$Z == 0]==1)

#compute the proportion of never-takers
never_takers <- mean(data$D[data$Z == 1]==0)

#compute the proportion of compliers
```

```r
compliers <- 1 - always_takers - never_takers

#make sure proportions sum to 1 and print to console
sum <- always_takers + never_takers + compliers

#show sum in console
print(paste("Sum of proportions:", sum))
```

```
[1] "Sum of proportions: 1"
```

ii.

We are not computing the proportion of defiers because we are assuming monotonicity, which rules out the existence of defiers. Monotonicity means that there are no units who would take the treatment when assigned to control and refuse it when assigned to treatment. Therefore, under this assumption, the proportion of defiers is zero by definition, and we focus on classifying units as always-takers, never-takers, or compliers.

Part B

```r
#Estimate the ITT, the LATE using the Wald estimator, and the LATE using the
#2SLS estimator (without covariates) and print the results to console.

#estimate the ITT
estimated_ITT <- mean(data$Y[data$Z == 1]) - mean(data$Y[data$Z == 0])

#estimate the LATE using the Wald estimator
LATE_wald <- (
  mean(data$Y[data$Z == 1]) - mean(data$Y[data$Z == 0])
) / (
  mean(data$D[data$Z == 1]) - mean(data$D[data$Z == 0])
)

#estimate the LATE using the 2SLS estimator (without covariates)
# create dataframe for model fitting
df <- cbind.data.frame(Y = data$Y, D = data$D, Z = data$Z)

# fit iv model
ivreg_diag <- ivDiag(
  Y = "Y",
```

```r
  D = "D",
  Z = "Z",
  data = df,
  parallel = FALSE
)
```

```
Bootstrapping:

Bootstrap took25.413sec.

AR Test Inversion...
```

```r
# print estimates from model to console
names(ivreg_diag)
```

```
 [1] "est_ols"  "est_2sls" "AR"       "F_stat"   "rho"      "tF"
 [7] "est_rf"   "est_fs"   "p_iv"     "N"        "N_cl"     "df"
[13] "nvalues"
```

```r
ivreg_diag$est_2sls
```

```
           Coef     SE      t CI 2.5% CI 97.5% p.value
Analytic 0.0819 0.0289 2.8316  0.0252   0.1386  0.0046
Boot.c   0.0819 0.0281 2.9178  0.0304   0.1380  0.0040
Boot.t   0.0819 0.0289 2.8316  0.0270   0.1368  0.0020
```

Part C

```r
#Evaluate the strength of the instrumental variable using the first-stage F-test and
#the Anderson-Rubin test. Print the results to console and interpret your findings.

# 1st-stage F-statistics
ivreg_diag$F_stat
```

```
 F.standard    F.robust   F.cluster F.bootstrap F.effective
   4873.659    4342.653          NA    4253.535    4342.653
```

```r
# Anderson-Rubin weak instruments test
ivreg_diag$AR
```

```
$Fstat
        F       df1       df2         p
   8.0460    1.0000 1602.0000    0.0046

$ci.print
[1] "[0.0258, 0.1386]"

$ci
[1] 0.0258 0.1386

$bounded
[1] TRUE
```

The F-statistic value of over 4000 indicates that we have a very strong instrument, as any F-statistic value higher than 10 is conventionally considered sufficient to satisfy the relevance restriction. The Anderson–Rubin test produces a confidence interval that excludes zero, further confirming that the estimated treatment effect is statistically significant and that the instrument is not weak.

Question 4

```r
# Load data
data3 <- read.csv("data3-1.csv")
```

Part A

ii. *Let's impose the following assumption of monotonicity* If $D_i(1) \neq D_i(0)$, then $D_i(1) = D_i(2)$. *Explain what this assumption states.*

Monotonicity is the assumption of "no defiers"–that if someone is randomly assigned treatment, they either comply because of treatment effect or because they are an always-taker and would have anyways. Treatment cannot induce people to "defy" and refuse treatment. This statement of monotonicity in our example means that if assigned to treatment for the community school, students either comply and attend community school, attend community school because they are an "always community school" unit and would have anyways, or remain in the school level they would have under control. No student can "downgrade" their school level with treatment

— if they would have attended community school under control, they cannot move down to public school or staying home under treatment.

ii. Present the *remaining* principal strata in a table.

Treatment status	Zi = 0	Zi = 1	Label
	0	0	Never taker
	0	2	Home-to-community
	1	1	Always public
	1	2	Public-to-community
	2	2	Always community

Part B

i. Complete the table with the principal strata that comprise the respective groups formed by crossing D_i and Z_i

Treatment Status	Zi = 0	Zi = 1
Di = 0	(0,0)	(0,0)
Di = 1	(1,1) (1,2)	(1,1)
Di = 2	(2,2)	(2,2) (0,2) (1,2)

ii.

```r
prop.table(table(data3$D, data3$Z), margin = 2)
```

```
            0         1
  0 0.3722222 0.1976428
  1 0.4750000 0.2901179
  2 0.1527778 0.5122393
```

Part C

ii. Calculate the proportion of each principal stratum. Make sure that these proportions sum to 1 and print the results to console.

First, the never-takers are the *only* stratum in Di=0|Zi=1, so we can find them easily.

```r
p_D0_Z1 <- prop.table(table(data3$D, data3$Z), margin = 2)["0", "1"]
# pulling from the table already made
p00 <- p_D0_Z1
p00
```

```
[1] 0.1976428
```

Second, the always-public schoolers are the only stratum in Di=1|Zi=1, so we can find them easily.

```r
p_D1_Z1 <- prop.table(table(data3$D, data3$Z), margin = 2)["1", "1"]
p11 <- p_D1_Z1
p11
```

```
[1] 0.2901179
```

Third, the always-community-school people are the only stratum found in Di=2|Zi=0, so we can find their proportion easily.

```r
p_D2_Z0 <- prop.table(table(data3$D, data3$Z), margin=2)["2", "0"]
p22 <- p_D2_Z0
p22
```

```
[1] 0.1527778
```

With random assignment to treatment, we can take the proportion of the sample that is always-community-schoolers in Di=2|Zi=0 is equivalent to the proportion in Di=2|Zi=1, which gets us closer to identifying both of our complier proportions.

There are two types of compliers, those who would have stayed at home but, under treatment, attend community school (home-to-community) and those who would have been in public school but, under treatment, enter community schools (public-to-community).

To get to them, I am going to need the remaining cells:

```r
p_D1_Z0 <- prop.table(table(data3$D, data3$Z), margin=2)["1","0"]
p_D2_Z1 <- prop.table(table(data3$D, data3$Z), margin=2)["2","1"]
```

For (0,2), our would have stayed at home but then attend community school if treated group, we see that they only exist in cell Di=2|Zi=1 but in a mix with (2,2) and (1,2). We have the proportion for (2,2), but need to find (1,2) to be able to isolate (0,2).

Group (1,2)–would stay in public school under control but move to community school when treated—appears in both Di=1|Zi=0 and Di=2|Zi=1, but they are only with (1,1) in the former cell. So, we use

$$p_{1,2} = p_{D=1|Z=0} - p_{1,1}$$

```
p12 <- p_D1_Z0 - p11
p12
```

```
[1] 0.1848821
```

Now we can look at stratum (0,2):

$$p_{0,2} = p_{D=2|Z=1} - p_{1,2} - p_{2,2}$$

```
p02 <- p_D2_Z1 - p22 - p12
p02
```

```
[1] 0.1745794
```

We have now found the proportion of each principal stratum. Now we check that they add to 1:

```
p00 + p11 + p22 + p12 + p02
```

```
[1] 1
```

Part D

Under the assumption of monotonicity, the local average treatment effect (LATE) represents the effect on respondents whose choices are affected by the instrument. Calculate the total proportion of such respondents based on your answer in 4c. Estimate the LATE as the ratio between the ITT estimate and this proportion.

The respondents whose choices are affected by the instrument are those stratum that change their behavior with treatment, so that eliminates the never-takers (0,0) the always public schoolers (1,1), and the always community schoolers (2,2).

```r
pi_c <- p12 + p02
pi_c
```

```
[1] 0.3594616
```

The ITT is the average outcome among the treated minus the average outcome among control, regardless of compliance:

$$ITT = E[Y|Z = 1] - E[Y|Z = 0]$$

So, we calculate that by

```r
Y1 <- mean(data3$Y[data3$Z == 1])
Y0 <- mean(data3$Y[data3$Z ==0])

ITT <- Y1 - Y0
ITT
```

```
[1] 0.2969116
```

To estimate the LATE, we compute the ratio between the ITT estimate and the compliance rate:

```r
LATE <- ITT / pi_c
LATE
```

```
[1] 0.8259898
```